Philip Crosby [Cro79], in his landmark book on quality, provides a wry answer to this question:

> The problem of quality management is not what people don't know about it. The problem is what they think they do know . . .
>
> In this regard, quality has much in common with sex. Everybody is for it. (Under certain conditions, of course.) Everyone feels they understand it. (Even though they wouldn't want to explain it.) Everyone thinks execution is only a matter of following natural inclinations. (After all, we do get along somehow.) And, of course, most people feel that problems in these areas are caused by other people. (If only they would take the time to do things right.)

Indeed, quality is a challenging concept—one that we addressed in some detail in Chapter 15.[1]

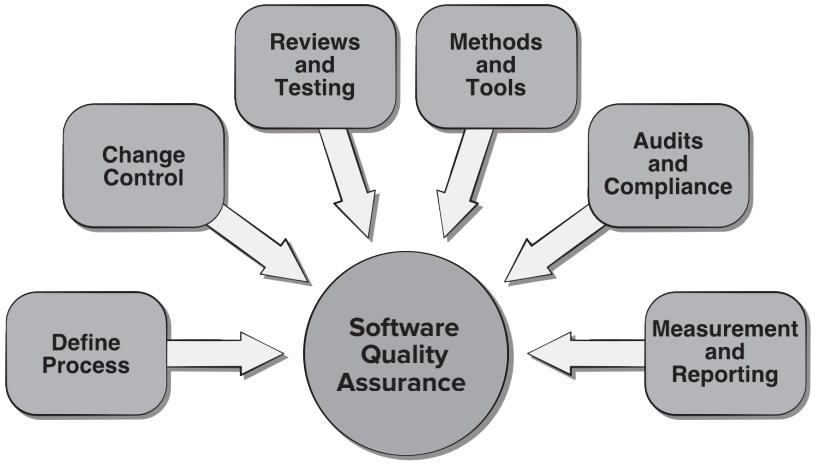
Some software developers continue to believe that software quality is something you begin to worry about after code has been generated. Nothing could be further from the truth! *Software quality assurance* (often called *quality management*) is an umbrella activity (Chapter 2) that is applied throughout the software process.

Software quality assurance (SQA) encompasses (Figure 17.1): (1) an SQA process, (2) specific quality assurance and quality control tasks (including technical reviews and a multitiered testing strategy), (3) effective software engineering practice (methods and tools), (4) control of all software work products and the changes made to them (Chapter 22), (5) a procedure to ensure compliance with software development standards (when applicable), and (6) measurement and reporting mechanisms.

In this chapter, we focus on the management issues and the process-specific activities that enable a software organization to ensure that it does "the right things at the right time in the right way."

FIGURE 17.1

Software quality assurance



1 If you have not read Chapter 15, you should do so now.

17.1 Background Issues

Quality control and assurance are essential activities for any business that produces products to be used by others. Prior to the twentieth century, quality control was the sole responsibility of the craftsperson who built a product. As time passed and mass production techniques became commonplace, quality control became an activity performed by people other than the ones who built the product.

The first formal quality assurance and control function was introduced at Bell Labs in 1916 and spread rapidly throughout the manufacturing world. During the 1940s, more formal approaches to quality control were suggested. These relied on measurement and continuous process improvement [Dem86] as key elements of quality management.

The history of quality assurance in software development parallels the history of quality in hardware manufacturing. During the early days of computing (1950s and 1960s), quality was the sole responsibility of the programmer. Standards for quality assurance for software were introduced in military contract software development during the 1970s and have spread rapidly into software development in the commercial world [IEE17]. Extending the definition presented earlier, software quality assurance is a "planned and systematic pattern of actions" [Sch01] that are required to ensure high quality in software. The scope of quality assurance responsibility might best be characterized by paraphrasing a once-popular automobile commercial: "Quality Is Job #1." The implication for software is that many different constituencies have software quality assurance responsibility—software engineers, project managers, customers, salespeople, and the individuals who serve within an SQA group.

The SQA group serves as the customer's in-house representative. That is, the people who perform SQA must look at the software from the customer's point of view. Does the software adequately meet the quality factors noted in Chapter 15? Have software engineering practices been conducted according to preestablished standards? Have technical disciplines properly performed their roles as part of the SQA activity? The SQA group attempts to answer these and other questions to ensure that software quality is maintained.

17.2 Elements of Software Quality Assurance

Software quality assurance encompasses a broad range of concerns and activities that focus on the management of software quality. These can be summarized in the following manner [Hor03]:

Standards. The IEEE, ISO, and other standards organizations have produced a broad array of software engineering standards and related documents. Standards may be adopted voluntarily by a software engineering organization or imposed by the customer or other stakeholders. The job of SQA is to ensure that standards that have been adopted are followed and that all work products conform to them.

Reviews and audits. Technical reviews are a quality control activity performed by software engineers for software engineers (Chapter 16). Their

intent is to uncover errors. Audits are a type of review performed by SQA personnel with the intent of ensuring that quality guidelines are being followed for software engineering work. For example, an audit of the review process might be conducted to ensure that reviews are being performed in a manner that will lead to the highest likelihood of uncovering errors.

Testing. Software testing (Chapters 19 through 21) is a quality control function that has one primary goal—to find errors. The job of SQA is to ensure that testing is properly planned and efficiently conducted so that it has the highest likelihood of achieving its primary goal.

Error/defect collection and analysis. The only way to improve is to measure how you're doing. SQA collects and analyzes error and defect data to better understand how errors are introduced and what software engineering activities are best suited to eliminating them.

Change management. Change is one of the most disruptive aspects of any software project. If it is not properly managed, change can lead to confusion, and confusion almost always leads to poor quality. SQA ensures that adequate change management practices (Chapter 22) have been instituted.

Education. Every software organization wants to improve its software engineering practices. A key contributor to improvement is education of software engineers, their managers, and other stakeholders. The SQA organization takes the lead in software process improvement (Chapter 28) and is a key proponent and sponsor of educational programs.

Vendor management. Three categories of software are acquired from external software vendors—*shrink-wrapped packages* (e.g., Microsoft Office), a *tailored shell* [Hor03] that provides a basic skeletal structure that is custom tailored to the needs of a purchaser, and *contracted software* that is custom designed and constructed from specifications provided by the customer organization. The job of the SQA organization is to ensure that high-quality software results by suggesting specific quality practices that the vendor should follow (when possible) and incorporating quality mandates as part of any contract with an external vendor.

Security management. With the increase in cyber crime and new government regulations regarding privacy, every software organization should institute policies that protect data at all levels, establish firewall protection for mobile apps, and ensure that software has not been tampered with internally. SQA ensures that appropriate process and technology are used to achieve software security (Chapter 18).

Safety. Because software is almost always a pivotal component of human-rated systems (e.g., automotive or aircraft applications), the impact of hidden defects can be catastrophic. SQA may be responsible for assessing the impact of software failure and for initiating those steps required to reduce risk.

Risk management. Although the analysis and mitigation of risk (Chapter 26) is the concern of software engineers, the SQA organization ensures that risk management activities are properly conducted and that risk-related contingency plans have been established.

In addition to each of these concerns and activities, SQA works to ensure that software support activities (e.g., maintenance, help lines, documentation, and manuals) are conducted or produced with quality as a dominant concern.

17.3 SQA Processes and Product Characteristics

As we begin a discussion of software quality assurance, it's important to note that SQA procedures and approaches that work in one software environment may not work as well in another. Even within a company that adopts a consistent approach[2] to software engineering, different software products may exhibit different levels of quality [Par11].

The solution to this dilemma is to understand the specific quality requirements for a software product and then select the process and specific SQA actions and tasks that will be used to meet those requirements. The Software Engineering Institute's CMMI and ISO 9000 standards are the most commonly used process frameworks. Each proposes "a syntax and semantics" [Par11] that will lead to the implementation of software engineering practices that improve product quality. Rather than instantiating either framework in its entirety, a software organization can "harmonize" the two models by selecting elements of both frameworks and matching them to the quality requirements of an individual product.

17.4 SQA Tasks, Goals, and Metrics

Software quality assurance is composed of a variety of tasks associated with two different constituencies—the software engineers who do technical work and an SQA group that has responsibility for quality assurance planning, oversight, record keeping, analysis, and reporting.

Modern software quality assurance is often data driven, as shown in Figure 17.2. The product stakeholders define goals and quality measures, problem areas are identified, indicators are measured, and a determination is made as to whether or not process changes are needed. Software engineers address quality (and perform quality control activities) by applying solid technical methods and measures, conducting technical reviews, and performing well-planned software testing.

17.4.1 SQA Tasks

The charter of the SQA group is to assist the software team in achieving a high-quality end product. The Software Engineering Institute recommends a set of SQA activities that address quality assurance planning, oversight, record keeping, analysis, and reporting. These activities are performed (or facilitated) by an independent SQA group that:

Prepares an SQA plan for a project. The plan is developed as part of project planning and is reviewed by all stakeholders. Quality assurance activities performed by the software engineering team and the SQA group are governed by the plan. The plan identifies evaluations to be performed, audits and

2 For example, CMMI-defined process and practices (Chapter 28).

FIGURE 17.2
Software quality assurance



reviews to be conducted, standards that are applicable to the project, procedures for error reporting and tracking, work products that are produced by the SQA group, and feedback that will be provided to the software team.

Participates in the development of the project's software process description. The software team selects a process for the work to be performed. The SQA group reviews the process description for compliance with organizational policy, internal software standards, externally imposed standards (e.g., ISO-9001), and other parts of the software project plan.

Reviews software engineering activities to verify compliance with the defined software process. The SQA group identifies, documents, and tracks deviations from the process and verifies that corrections have been made.

Audits designated software work products to verify compliance with those defined as part of the software process. The SQA group reviews selected work products; identifies, documents, and tracks deviations; verifies that corrections have been made; and periodically reports the results of its work to the project manager.

Ensures that deviations in software work and work products are documented and handled according to a documented procedure. Deviations may be encountered in the project plan, process description, applicable standards, or software engineering work products.

Records any noncompliance, and reports to senior management. Noncompliance items are tracked until they are resolved.

In addition to these activities, the SQA group coordinates the control and management of change (Chapter 22) and helps to collect and analyze software metrics.

SAFEHOME



Software Quality Assurance

The scene: Doug Miller's office as the *SafeHome* software project begins.

The players: Doug Miller, manager of the *SafeHome* software engineering team, and other members of the product software engineering team.

The conversation:

Doug: How are things going with the informal reviews?

Jamie: We're conducting informal reviews of the critical project elements in pairs as we code but before testing. It's going faster than I thought.

Doug: That's good, but I want to have Bridget Thorton's SQA group conduct audits of our work products to ensure that we're following our processes and meeting our quality goals.

Venod: Aren't they already doing the bulk of the testing?

Doug: Yes, they are. But QA is more than testing. We need to be sure that our documents are evolving along with our code and that we're making sure we don't introduce errors as we integrate new components.

Jamie: I really don't want to be evaluated based on their findings.

Doug: No worries. The audits are focused on conformance of our work products to the requirements and the process activities we've defined. We'll only be using audit results to try to improve our processes as well as our software products.

Vinod: I must believe it's going to take more of our time.

Doug: In the long run it will save us time when we find defects earlier. It also costs less to fix defects if they're caught early.

Jamie: That sounds like a good thing then.

Doug: It's also important to identify the activities where defects were introduced and add review tasks to catch them in the future.

Vinod: That'll help us determine if we're sampling carefully enough with our review activities.

Doug: I think SQA activities will make us a better team in the long run.

17.4.2 Goals, Attributes, and Metrics

The SQA activities described in the preceding section are performed to achieve a set of pragmatic goals:

Requirements quality. The correctness, completeness, and consistency of the requirements model will have a strong influence on the quality of all work products that follow. SQA must ensure that the software team has properly reviewed the requirements model to achieve a high level of quality.

Design quality. Every element of the design model should be assessed by the software team to ensure that it exhibits high quality and that the design itself conforms to requirements. SQA looks for attributes of the design that are indicators of quality.

Code quality. Source code and related work products (e.g., other descriptive information) must conform to local coding standards and exhibit characteristics that will facilitate maintainability. SQA should isolate those attributes that allow a reasonable analysis of the quality of code.